

Pratik Jadhav

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SUMMARY

AI/ML Engineer building production LLM agents, multimodal AI systems, and evaluation/observability infrastructure. Reduced agent inference latency 77% (18s→4.2s), improved computer-vision defect detection from 34% to 92%, and shipped production AI services with FastAPI, Docker, GCP/Vertex AI, and MLflow

TECHNICAL SKILLS

Languages: Python, SQL

AI/ML: LLMs, RAG, DSPy, LangChain, LangGraph, Hugging Face, PyTorch, TensorFlow, Scikit-learn, CNNs, NLP, Computer Vision, LoRA

Backend & Data: FastAPI, REST APIs, Async Python, WebSockets, Celery, RabbitMQ, Redis, PostgreSQL, MongoDB, MySQL, FAISS, BM25, Qdrant

MLOps & Cloud: MLflow, Langfuse, Prometheus, Grafana, Vertex AI, Docker, TensorFlow Serving, GitHub Actions

PROFESSIONAL EXPERIENCE

AI/ML Engineer, Joshnik AI Labs

03/2025 – Present | Bengaluru, India

- Optimized a **brand-intelligence agent** serving **16K+ brands**, reducing latency 77% (**18s→4.2s**) through parallel retrieval, deterministic ranking, context reduction, and MLflow evaluation.
- Architected a **multimodal AI platform** for a **6,000+ participant event**, combining a **DSPy/ReAct campaign agent** with **MTCNN + InceptionResnetV1 face search** and WebSocket streaming across **~10K faces/campaign**.
- Engineered an **AI professional-discovery workflow** combining LLM reasoning, Sentence Transformer embeddings, and retrieval across **50K+ profiles, 48 categories, and 27 locations**.
- Established production **LLMOps** with DSPy, MLflow, Prometheus, and Redis Streams for evaluation, tracing, telemetry, and ReAct/tool-calling workflows.
- Productionized **restaurant GenAI workflows** using Claude 3.5 Sonnet for menu generation, consent drafting, deterministic routing, schema validation, AI image generation, and human-review gates across live venues.

Machine Learning Intern, Alemeno Private Limited

04/2024 – 10/2024 | Mumbai, India

- Engineered a **hybrid vision-LLM LCD defect detection system** across **17 defect classes**, combining LLM prompting, CNNs, object detection, and preprocessing to improve accuracy **34%→92%**.
- Curated **10K+ annotations** and built preprocessing and hard-negative mining pipelines, improving difficult-class detection **54%→81%** while training models on Vertex AI.
- Engineered a **satellite ship-detection pipeline** across **42K+ SAR/EO images** using RF-DETR, image tiling, automated annotation, COCO conversion, and resumable batch processing.

Applied AI Intern, Atria University

08/2023 – 03/2024 | Bengaluru, India

- Built a **RAG-based agricultural assistant** using large-scale document retrieval, RAG/RAFT experimentation, and a custom **LLM-as-a-Judge evaluation framework**, improving retrieval precision **25–35%** while reducing hallucinations.

Machine Learning Intern, Cheslab Private Limited

04/2023 – 06/2023 | Bengaluru, India

- Developed **CNN-based skin disease classification pipelines** using **100K+ medical images**, contributing to dataset engineering, preprocessing, class balancing, model training, and healthcare AI application integration.

PROJECTS

Curalink: AI Medical Research Assistant 

3rd Place, AI Hackathon (800+ Participants) · FastAPI, Flan-T5, LoRA, Groq Llama-3.3-70B, MiniLM, MongoDB

- Fine-tuned **Flan-T5-base with LoRA/PEFT** on **8K medical training examples**, adding task-specific query-expansion examples for structured PubMed search; merged and deployed the adapted model to **Hugging Face**.
- Architected a **multi-source medical research RAG pipeline** concurrently retrieving and deduplicating evidence from **PubMed, OpenAlex, and ClinicalTrials.gov**, combining semantic similarity, recency, and source credibility to rank evidence before grounded **Llama-3.3-70B** synthesis.

PDF Corpus Chat: Multi-Document RAG System 

Python · Streamlit · LangChain · Qdrant · Gemini · BM25 · Cross-Encoder · Langfuse · Docker · GitHub Actions

- Architected a **multi-document RAG system** with isolated Qdrant stores, **hybrid dense + BM25 retrieval, RRF, and cross-encoder reranking**; reduced hallucination rate **38%** and achieved **<1.2s latency** through retrieval optimization and caching.
- Built an **automated RAG evaluation and CI pipeline** with **50 verified Q&A cases**, LLM-as-a-Judge metrics, citation validation, Langfuse tracing, and **165+ automated tests**, enforcing quality thresholds on every deployment.

EDUCATION

Atria University, B.Tech, Computer Science (Digital Transformation)

11/2021 – 08/2025 | Bengaluru, India

- CGPA: 8.53/10.0**